

AERIS Expert Labeling User Guide

DRAFT - for Dr. Bracco's review; not final until confirmed in writing

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This guide explains how to review a blinded AERIS claim packet. The example packet you have is for format review only. Nothing becomes an official label until my pre-freeze checks pass and you've confirmed this guide in writing.

What's in the packet

Each case starts with the anomaly itself: source, metric, observed and expected values, UTC time, location, severity, and which detectors flagged it. After that comes a condensed table of the stored 72-hour evidence, a map of the ground stations, wind panels, and then the claims. Each case ends with one optional box: your best guess at the most likely true cause.

Duplicate claims are printed once (different models often write the same claim word for word). The claims are shuffled (the shuffle is fixed per labeler, so your packet and mine differ) and each one carries a presentation number. The packet never shows which model wrote a claim, the model's confidence, or any of my automated scores, so nothing can nudge your judgment toward or away from what the pipeline thinks.

The data sources, briefly

You'll see these names on the plots and in the evidence table:

- OpenAQ: an aggregator of government air quality monitors worldwide. I only include stations I could verify as real regulatory monitors.
- TCEQ: Texas state monitoring stations (NO₂, SO₂, CO), read from the commission's public site.
- PurpleAir: cheap optical PM_{2.5} sensors that people mount on their houses. Dense coverage, noisier data; they pass through their own quality screen.
- ASOS: the automated weather stations at airports. Direct anemometer and thermometer readings.
- OpenWeather: a commercial weather product that blends several inputs, including station data.
- NOAA GFS: the global forecast model. I use its 6000 analysis fields (winds, boundary layer height); it assimilates surface observations, so it is not independent of ASOS.
- Sentinel-5P: the ESA satellite that measures column amounts of NO₂, SO₂, and other gases from orbit.
- EPA AQS: EPA's official historical archive of ground chemistry (NO₂, SO₂, CO). Used for the historical window only, not live collection.

Label definitions

Mark each claim you answer with exactly one of these:

- **Valid (V):** the claim is scientifically defensible at the precision stated and is supported by the evidence shown.
- **Invalid (I):** the evidence shown contradicts the claim, the claim misstates a measurement, or the physical reasoning is not defensible.
- **Unsure (U):** the displayed evidence is insufficient, the measurements cannot resolve the claim, the wording is ambiguous, or the claim falls outside what you can judge confidently.

Two consequences worth calling out:

- Missing or thin evidence means **Unsure**, not Invalid.
- A cause that is plausible but unproven stays **Unsure** unless the evidence actually contradicts it.

These definitions are proposals until you confirm them in writing.

How to mark claims

1. Mark exactly one box (V, I, or U) for each claim you answer.
2. If you skip a claim, leave all three boxes blank. I record a blank as missing; I never quietly turn it into Unsure.
3. Add a short note on Invalid or Unsure when the reason isn't obvious. Notes on Valid are optional.
4. If a claim is compound or ambiguous, or leans on something it never states, mark Unsure and say why in a few words. Claims are never silently dropped.
5. The event-level “most likely true cause” box is optional, and “insufficient evidence to determine a single cause” is a perfectly good answer.

Reading the plots

The station map: the black star is the anomaly, and the colored dots are every station that contributed data in the stored 72-hour window. The legend gives the count per source. There is no basemap on purpose; you see nothing the models didn't see.

The wind panels: one panel per wind channel (ASOS, GFS, OpenWeather). Each arrow is that station's wind observation nearest the event time, pointing downwind. Longer arrow means faster wind, but each panel has its own scale, so compare lengths within a panel, not across panels. The label above each panel gives the speed of its longest arrow. ASOS and OpenWeather arrows come from stored speed and direction values; GFS arrows come from its stored eastward and northward wind components.

Everything plotted comes from the same stored context the models saw, and an automated audit verifies that before any packet goes out.

Calm-wind flags

Sometimes the packet flags a wind-direction claim as calm. That happens when at least one relevant source reports an event-time wind speed below that source's cutoff. Transport claims check GFS, OpenWeather, and ASOS; point-source claims check GFS and OpenWeather. The flag lists the affected sources with their event speeds, their cutoffs, and the number of wind observations behind each cutoff. A flagged claim defaults to Unsure with the standard note "wind direction unstable under calm conditions." You never have to compute anything yourself.

The proposed cutoff is $\max(\text{mean} - 2 \times \text{SD}, 1.5 \text{ m/s})$, computed per channel from that channel's own wind speeds over the stored 72-hour window (SD is the population standard deviation). A claim is flagged only when the event-nearest speed is strictly below the cutoff. The 1.5 m/s floor is **proposed and waiting on your written confirmation**. It is not a final rule, and the final guide will state whatever value you confirm.

Missing or stale speed data never creates a calm flag. It just falls under the normal insufficient-evidence-means-Unsure rule.

Returning the packet

The annotated PDF you send back is the official record; I keep the file exactly as you return it. If one of your marks is ambiguous, I don't interpret it. I'll email you a short numbered list of the specific items and wait for your answer. Claims you skipped stay recorded as missing, exactly as marked.

Versioning

Once official labeling starts, this guide is frozen. If any definition, threshold, annotation rule, or any step in how I record your marks has to change after that, everything stops: I version the guide, disclose the change to you, and run a pre-declared sensitivity analysis to show whether the change affected any label.